LAB 7: Fuzzing for Input Validation Bugs

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Experiment summary

This lab uses Hypothesis property-based testing to discover malformed and edge-case inputs for average_marks, parse_srn and short_name. The original implementation is retained separately to reproduce defects; student_grades.py contains the hardened implementation.

Files included

student_grades_buggy.py — original assigned program; test_student_grades_buggy.py — tests that expose original defects; student_grades.py — fixed program; test_student_grades.py — fuzz tests for the fixed program; README.md — setup and run commands.

Validation rules used

Marks: comma-separated integers from 0 to 100; empty values and invalid/out-of-range marks are rejected. SRN: follows the 13-character sample format PES1UG23CS123 and returns campus, year and department. Name: requires at least two whitespace-separated tokens and uses the first and last tokens.

Run commands (Windows PowerShell)

```
py -m venv .venv
.\.venv\Scripts\Activate.ps1
py -m pip install -r requirements.txt
py student_grades_buggy.py
py -m pytest -q test_student_grades_buggy.py
py student_grades.py
py -m pytest -q test_student_grades.py
```

Screenshot evidence to add before submission

1. Original program run (normal sample inputs). 2. Failing output from test_student_grades_buggy.py. 3. Fixed program run. 4. Passing output from test_student_grades.py. Capture these from your own VS Code terminal after installing dependencies; screenshots are intentionally not fabricated.

Important: the PDF is a report template with the experiment details. Insert your four genuine terminal screenshots before submitting it as the final screenshot evidence.